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Gait and Function after Total Hip Arthroplasty: A Systematic Review and Meta-Analysis of the Hardinge versus Posterolateral Approaches

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Abstract

Background: This systematic review and meta-analysis aimed to compare postoperative gait and functional outcomes between the Hardinge approach (HA) and

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the posterolateral approach (PLA) for total hip arthroplasty (THA).

Methods: A comprehensive literature search was conducted in PubMed and Embase for studies directly comparing HA and PLA. Eligible studies were selected based on predefined inclusion and exclusion criteria. Two reviewers independently performed study screening, quality assessment, and data extraction.

Results: Twenty-six studies, including 245,620 patients (HA: n=88,193; PLA: n=157,427), were included. Compared to the HA group, the PLA group was associated with better gait velocity and abductor muscle strength, while the HA group exhibited a lower hip dislocation rate but higher risks of heterotopic ossification and infection. No significant intergroup differences were found in the Harris Hip Score, stride length, ground reaction force, periprosthetic fracture, or radiographic outcomes.

Conclusions: This meta-analysis suggests that PLA offers advantages in gait velocity and abductor power recovery, while HA is associated with a lower dislocation rate. This potential benefit of HA must be balanced against its observed higher risks of heterotopic ossification and surgical site infection. These findings, however, are derived from observational studies with inherent confounding factors, such as variations in implant design and surgical technique. Therefore, the reported outcome differences may not be attributable solely to the surgical approach.

Trial registration: Registration ID, CRD42025634164

Clinical trial number: not applicable.

Keywords: Total hip arthroplasty, Hardinge approach, Posterolateral approach, Meta-analysis

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Background

Total hip arthroplasty (THA) has established itself as the definitive surgical intervention for end-stage hip arthritis, consistently producing excellent results in terms of pain alleviation and functional recovery. Widely regarded as one of the most successful surgical interventions of the 20th century, the procedure continues to advance through enhancements in surgical techniques and rising patient expectations for optimal postoperative outcomes. In contemporary practice, the selection of a surgical approach plays a crucial role in determining the quality of the outcome, with each technique presenting distinct anatomical considerations and potential complications. Among the various approaches, the posterolateral approach remains

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widely employed due to its extensive surgical exposure and technical familiarity among orthopedic surgeons. However, it is also associated with an increased risk of prosthetic dislocation Error: Reference source not found. Conversely, the Hardinge approach (direct lateral approach) carries a risk of superior gluteal nerve injury, which can lead to postoperative abductor weakness and gait impairment due to muscle damage [3,4]. Clinical series have documented complications as persistent abductor deficiency and abnormal gait patterns.

Recent reviews [5-9], have synthesized evidence on various outcomes, yet a dedicated synthesis with a focus on quantitative gait parameters and abductor muscle strength, both of which are key indicators of functional performance, remains lacking. Furthermore, existing syntheses often do not incorporate a systematic assessment of the overall certainty of the accumulated evidence, which is crucial for contextualizing the findings and guiding clinical interpretation. To address the situation, we conducted this systematic review and meta-analysis with a dedicated focus on comparing gait and function between HA and PLA. Importantly, we applied the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) framework to evaluate the certainty of evidence for outcomes. This combined focus on functional recovery and methodological rigor aims to provide a more nuanced and reliably graded evidence base. Our investigation aims to synthesize current evidence to inform surgical decision-making and ultimately enhance postoperative quality of life for patients.

Methods

Search strategies

A study protocol was developed a priori, specifying the planned search strategy,

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inclusion/exclusion criteria, and outcomes for analysis. This review is registered in the PROSPERO database and follows the PRISMA reporting guidelines (CRD42025634164).

Two electronic databases (PubMed and Embase) were systematically searched for English-language articles published from inception until August 31, 2025. The search strategy, which incorporated keywords and subject headings for core concepts (e.g., total hip arthroplasty, surgical approach, gait, postoperative outcomes), is provided in the Supplementary Material. Furthermore, the reference lists of all included studies were reviewed to identify any additional relevant publications.

Inclusion criteria

(1) Population: Patients receiving primary, unilateral THA.

(2) Intervention & Comparison: Direct comparison between the Hardinge (HA) and posterolateral (PLA) approaches

(3) At least one relevant outcome was reported, prioritized as follows:

- **Primary:** Gait (speed, stride length, ground reaction force), abductor power, Harris Hip Score (HHS).
- **Secondary:** Dislocation, Infection, Fracture, Heterotopic ossification (HO), radiographic outcome.

(4) **Study Design**: RCTs or high-quality comparative studies.

Exclusion criteria

1. Non-comparative studies.

2. Inaccessible or irrelevant data.

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3. Duplicate publications or reviews

Study selection

The study selection was performed independently by two professionally trained researchers. Initially, duplicate records were identified and removed using the deduplication feature in Zotero software. The subsequent screening process consisted of two stages: first, the titles and abstracts of the remaining records were screened against the eligibility criteria; second, the full texts of potentially relevant studies were retrieved and thoroughly assessed. Any disagreements between the reviewers at either stage were resolved through discussion or, when necessary, by arbitration from a third senior researcher. Studies that ultimately met all inclusion criteria were advanced to the data extraction.

Data extraction

Two reviewers independently extracted the data from all included studies. The extracted indexes encompassed:

Bibliometric and Demographic Data: author, country, mean age, sample size, sex, Body Mass Index (BMI), study type, and follow-up time.

Primary Outcomes: Gait (speed, stride length, ground reaction force), abductor power, HHS.

Secondary Outcomes: (1) Complications: loosening, infection, dislocation, fracture, and HO.

(2) Radiographic Outcome: abduction angle, anteversion angle.

Any differences in data extraction were reconciled through discussion, with unresolved issues escalated to a third investigator for final adjudication.

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Risk of Bias Assessment

The methodological quality and risk of bias of the included studies were assessed independently by two reviewers. Any discrepancies in judgment were resolved through discussion or, when necessary, by consultation with a third senior researcher.

For the RCTs, we employed the Cochrane Risk of Bias 2 (RoB 2) tool which evaluated bias across five domains: (1) Randomization process; (2) Deviations from intended interventions; (3) Missing outcome data; (4) Measurement of the outcome; (5) Selection of the reported result. Each domain was judged as "low risk," "some concerns," or "high risk" of bias, leading to an overall risk-of-bias judgment for each study.

For the cohort studies, we used the Newcastle-Ottawa Scale (NOS) to assess quality. The NOS evaluated studies across three broad categories: (1) Selection (including Representative-ness, selection of the non-exposed, ascertainment of expose, and outcome not present at start); (2) Comparability (controlling for the most important factor and any additional factor); (3) Outcome assessment (including assessment method, duration, and adequacy of follow-up). A star system is used, with a maximum of 9 stars indicating the highest quality.

The results of risk of bias and quality assessments were considered when interpreting the findings of the meta-analysis. Specifically, they informed the discussion of the limitations of the included evidence and guided the interpretation of the pooled results, acknowledging that the conclusions are primarily based on observational data with inherent risks of confounding.

Statistical analysis

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Data were analyzed with Cochrane's Review Manager (RevMan, version 5.4.0; The Cochrane Collaboration). We calculated risk ratios (RR) for dichotomous variables and mean differences (MD) for continuous variables, both presented with 95% confidence intervals (CIs). Statistical significance for pooled effects was defined as $P < 0.05$. Study heterogeneity was quantified with the I^2 statistic, with values over 50% (coupled with a $P < 0.10$ from the Chi-squared test) indicating substantial heterogeneity and warranting the use of a random-effects model; otherwise, a fixed-effect model was applied. The results are graphically displayed in forest plots.

Handling of Heterogeneity

We planned to investigate heterogeneity as part of our analytic strategy. When substantial statistical heterogeneity was identified, we used the following analyses to test the robustness of the primary findings:

1.Sensitivity Analysis: A leave-one-out sensitivity analysis was performed by iteratively removing each study to assess its individual impact on the pooled estimate and the I^2 statistic.

2.Subgroup Analysis: We conducted subgroup analyses based on clinically relevant factors hypothesized to influence outcomes.

Assessment of the Certainty of Evidence

The overall certainty of evidence for each pre-specified critical and important outcome was assessed using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach. Two reviewers independently rated the certainty of evidence. Any discrepancies were resolved through discussion or, if necessary, by consultation with a third senior author. The certainty rating for outcomes based on the following five domains: (1) Risk of bias;(2) Inconsistency;

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(3) Indirectness; (4) Imprecision;(5) Publication bias. The judgments and reasons for upgrading or downgrading the evidence for each outcome are summarized in a 'Summary of Findings' (SoF) table, which was created using the GRADE profiler (version 3.6).

Results

Study characteristics and quality evaluation

A total of 1,027 records initially retrieved, 901 remained after de-duplication. Following a screening of titles/abstracts and subsequent full-text assessment, 29 studies (27 non-RCTs and 2 RCTs) met the inclusion criteria and were included in the meta-analysis. The PRISMA flow diagram (Fig. 1) outlines the exclusion reasons, which included systematic reviews, non-primary THA, and unextractable data.

The baseline characteristics of the 26 included studies are summarized in Table 1. These studies encompassed a total of 245,620 patients (HA: 88,193; PLA: 157,427). The sample sizes of individual studies ranged from 10 to 90,259, with follow-up durations varying from 1 to 72 months. Methodological quality was assessed using the NOS for the 24 non-randomized studies, with seven prospective and seventeen retrospective cohorts rated as high or moderate quality (Table 2). The risk of bias for the two RCTs was evaluated using the RoB 2 (Fig. 2, Fig. 3). The summary of findings for GRADE is presented in Table 3. The detailed judgements for each GRADE domain are provided in the Evidence Profiles (Supplementary Material).

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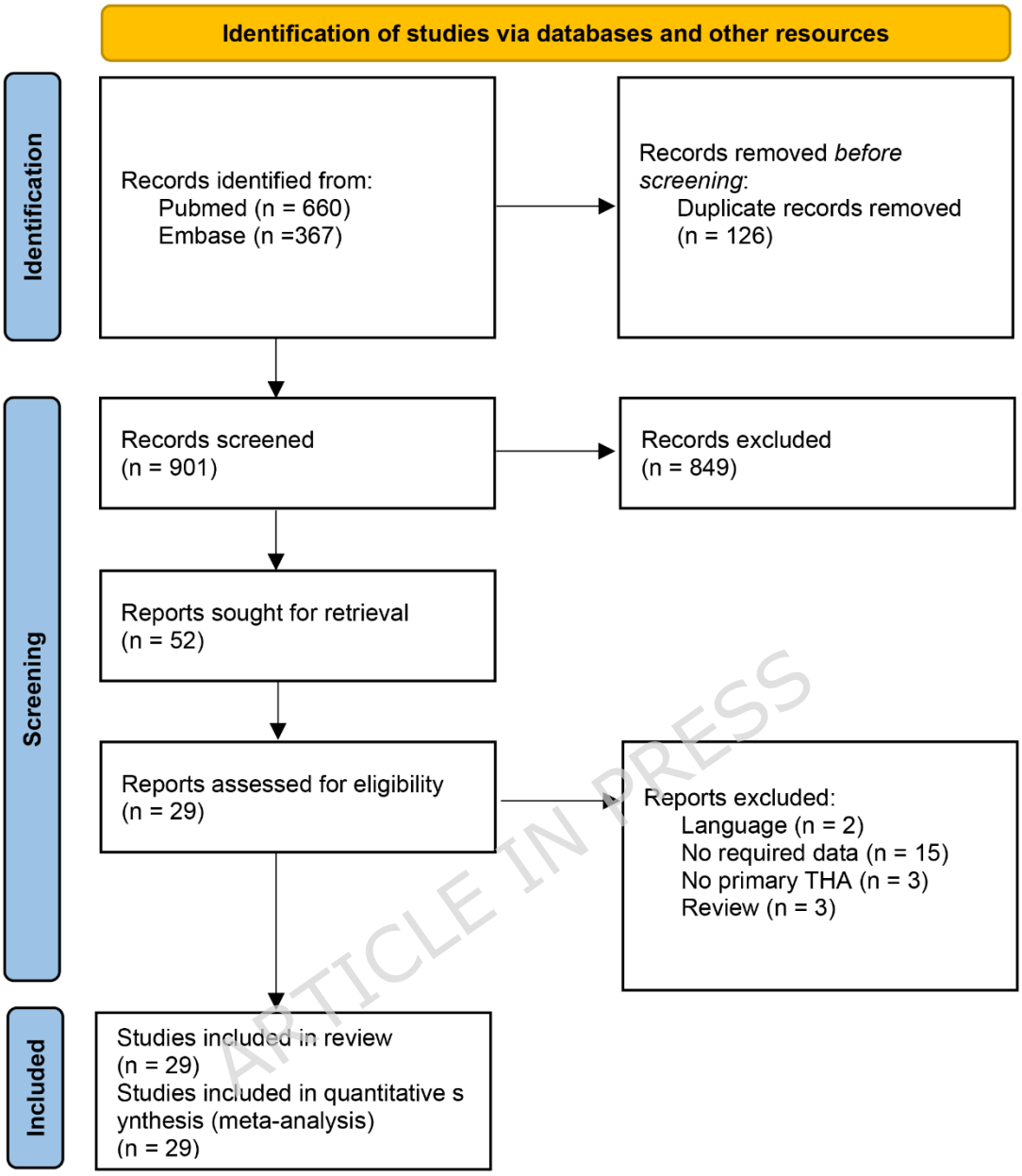

Fig. 1 Flow of study selection

Fig. 2 Risk of bias graph: review authors' judgments about each risk of bias item presented as percentages across all included studies

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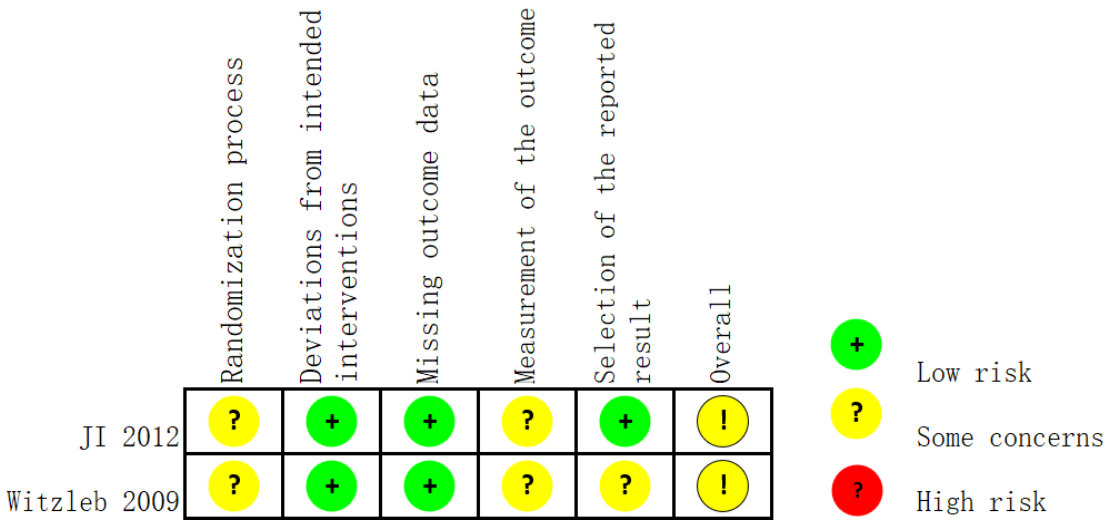

Fig. 3 Risk of bias summary for included studies.

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Table 1 Characteristics of included studies

Author	Country	Mean age (years)	HA/PLA No. of patients (n)	Male	BMI	Study	Mean follow-up (months)	Outcomes
Jeyaraman[10]	India	49.6	47/42	32/25	NC		Prospective	1,2,4,
Barber[11]	USA	70	21/28	8/14	NC		Retrospective	1,4,5,10
Xie[12]	China	62.6	120/201	48/91	24.7/24.9		Retrospective	1,7,8
Ritter[13]	USA	68	130/190	52/82	NC		Retrospective	1,2,6
Fleischman[14]	USA	63.4	8561/2160	4186/964	29.9/28.8		Retrospective	1,4
Liang[15]	China	41.2	21/20	4/9	25.56/25.67		Retrospective	1,2,4,5,6,7,8
van Erp[16]	Netherlands	65	169/127	48/42	27/26		Prospective	5
Tay[17]	Australia	66.6	618/870	NC	NC		Retrospective	1,2,3
Hermansen[18]	America	NC	1226/29879	NC	NC		Retrospective	1
Thompson[19]	Germany	64.8	100/100	NC	30/28.4		Retrospective	7
Rogmark[20]	Sweden	74.8	4832/3199	3422/232	NC		Prospective	1

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Castioni[21]	England	65.3	72/68	39/33	29.4/29.3	Retrospective	1,4,6
Skoogh[22]	Sweden	68.8	44933/51848	1860 4/22447	27.3/27.5	Retrospective	1,2,3
Amouzadeh Omrani[23]	Iran	62.3	25/25	NC	26.4/24.5	Retrospective	6
Gupta[24]	New Zealand	44.8	2499/2255	706/651	26.6/26.3	Retrospective	1,2,3,4
Hoskins[25]	Australia	NC	24468/65791	1135 1/3095 3	NC	Retrospective	1,2,3,4
Olivero[26]	Italy	61	35/277	15/162	NC	Retrospective	5
Kumar[27]	India	63.2	100/100	47/49	28.6/29.1	Retrospective	6
Witzleb[28]	Germany	58	30/30	15/15	26.6/28.9	RCT	1,2,3,6,7
JI[29]	Korea	52	97/99	58/54	24.3/24.3	RCT	3,7,8
Whatling[30]	England	64.21	5/5	NC	31.2/30.6	Retrospective	9
Queen[31]	America	58	8/12	4/6	27.7/25.2	Prospective	6,9
Queen[32]	America	61.25	12/18	NC	28.4/27.6	Prospective	6,9
Robbins[33]	Canada	68	21/21	10/15	26.6/27.8	Retrospective	9
Ansari[34]	India	45	22/20	9/10	24.09/24.03	Prospective	10
Zeni[35]	America	59	21/42	13/19	29/28.6	Prospective	10

NC not clear

1. dislocation; 2. infection; 3. fracture; 4. loosening; 5. heterotopic ossification 6. HHS; 7. abduction angle; 8. anteversion angle 9. gait 10. abductor power

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Table 2 The literatures quality of cohort studies were evaluated using NOS

	Selection				Comparability		Outcome		
Study	Representativeness	Selection of non-exposed	Ascertainment of expose	Outcome not present at start	Comparability on most important factors	Comparability on other risk factors	Assessment of outcome	Long enough follow-up (median≥1 year)	Adequacy (completeness) of follow-up
Jeyaraman	*	*	*	*	*	-	*	*	*
Barber	*	*	*	*	*	-	*	*	*
Xie	*	*	*	*	*	-	*	*	*
Ritter	*	*	*	*	-	-	*	*	*
Fleischman	*	*	*	*	-	-	*	*	*
Liang	*	*	*	*	*	-	*	-	*
van Erp	*	*	*	*	-	-	*	*	*
Tay	*	*	*	*	-	-	*	-	*
Hermansen	*	*	*	*	*	-	*	*	*
Thompson	*	*	*	*	*	-	*	-	-
Rogmark	*	*	*	*	*	-	*	*	*
Castioni	*	*	*	*	*	-	*	*	*
Skoogh	*	*	*	*	*	-	*	*	*
Amouzadeh Omrani	*	*	*	*	*	-	*	-	*
Gupta	*	*	*	*	*	-	*	*	*

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Hoskins	*	*	*	*	*	-	*	*	*
Olivero	*	*	*	*	-	-	*	*	*
Kumar	*	*	*	*	*	-	*	*	*
Whatling	*	*	*	*	-	-	*	-	*
Queen	*	*	*	*	*	*	*	-	*
Queen	*	*	*	-	*	*	*	*	*
Robbins	*	*	*	-	-	-	*	*	*
Ansari	*	*	*	*	*	-	*	-	*
Zeni	*	*	*	*	-	-	*	*	-

Table 3 Summary of Findings Table

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	Quality of the evidence (GRADE)		Comments
	Assumed risk	Corresponding risk				
	PLA	**HA**				
gait analysis		**0.02 lower** (0.05 lower to 0 higher)		204 (4 studies)	⊕⊖⊖⊖ **very low**[1,4]	
Abductor power		**0.21 lower** (0.41 to 0.02 lower)		196 (3 studies)	⊕⊖⊖⊖ **very low**[1]	
HHS score		**1.37 lower** (2.73 lower to 0.02 higher)		1398 (8 studies)	⊕⊖⊖⊖ **very low**[1,2,4]	
dislocation	**Study population**		**RR 0.41** (0.24 to 0.7)	244159 (14 studies)	⊕⊖⊖⊖ **very low**[1,3]	
	12 per 1000	**5 per 1000** (3 to 9)				
	Moderate					
	11 per 1000	**5 per 1000** (3 to 8)				
infection	**Study population**		**RR 1.19** (1.08 to 1.32)	193778 (8 studies)	⊕⊖⊖⊖	
	7 per 1000	**9 per 1000** (8 to 10)				
	Moderate					

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	7 per 1000	**8 per 1000** (8 to 9)			**very low**[1]
heterotopic ossification	**Study population**	**RR 1.59** (1.18 to 2.13)	457 (4 studies)	⊕⊖⊖⊖	
	237 per 1000	**377 per 1000** (280 to 505)		**very low**[1]	
	Moderate				
	199 per 1000	**316 per 1000** (235 to 424)			

*The basis for the **assumed risk** (e.g. the median control group risk across studies) is provided in footnotes. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).
CI: Confidence interval; **RR:** Risk Ratio

GRADE Working Group grades of evidence
High quality: Further research is very unlikely to change our confidence in the estimate of effect.
Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate.
Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate.
Very low quality: We are very uncertain about the estimate.

1 Risk of bias exists 2 50%＜I2＜75% 3 I^2≥75% 4 CI crossed the clinically significant threshold

Gait analysis

Gait were analyzed by stratifying the data into three distinct parameters: speed, stride length, and ground reaction force (GRF). Although no statistically significant differences were found in stride length or GRF, a consistent trend across the analyses indicated superior postoperative gait recovery in the PLA group compared to the HA group (Fig.4).

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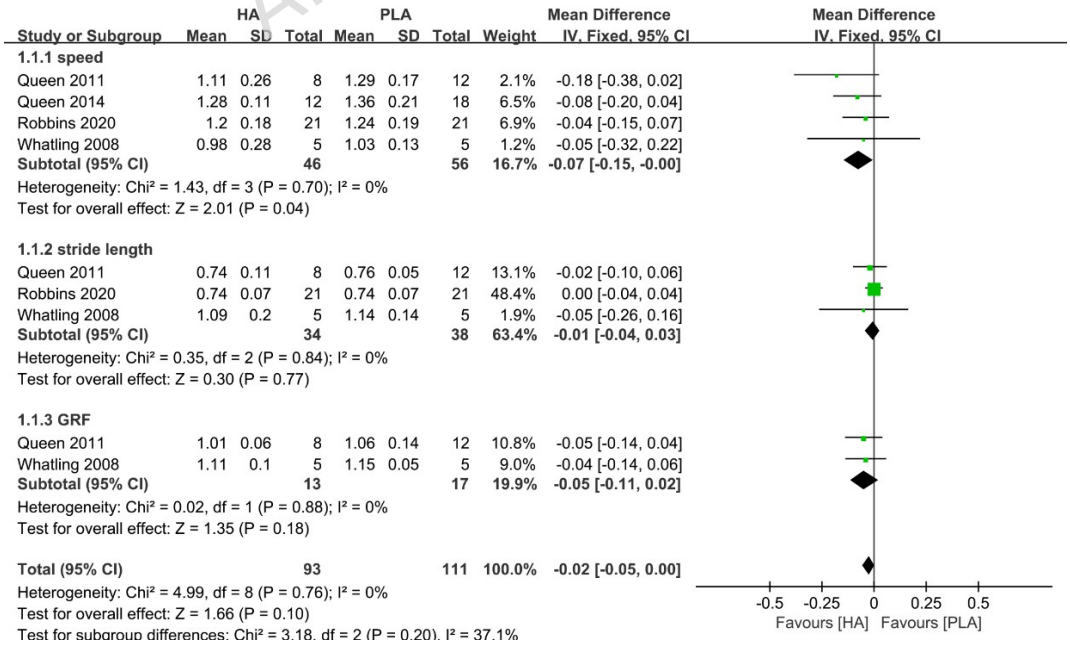

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Fig.4 Forest plot for gait analysis

Abductor power

Subgroup analysis of abductor power across three postoperative time points (3, 6, and 12 months) demonstrated a significant advantage for the PLA group over the HA group (MD = -0.21, 95% CI: -0.41 to -0.02; P = 0.03. Fig.5).

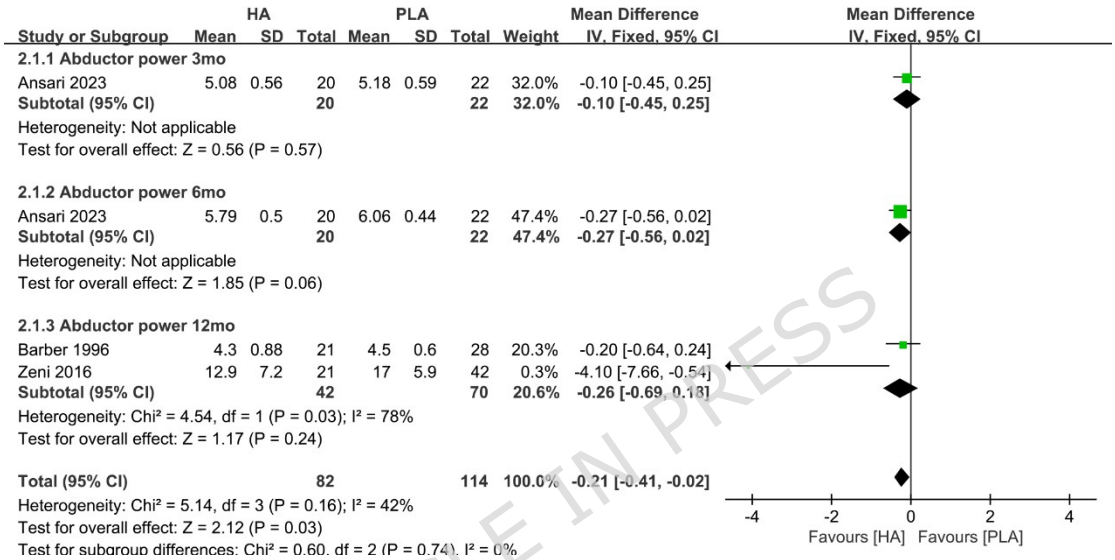

Fig.5 Forest plot for abductor power

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Dislocation

The rate of hip dislocation was significantly lower in HA group than in PLA group, based on data from 14 studies involving 244,159 participants [10-15,17,18,20-22,24,25,28] (Fig. 6). Subgroup analysis by data source revealed significant between-subgroup differences (Fig. 7). Among clinical studies, HA was associated with a significantly lower risk of dislocation compared to PLA (RR = 0.12, 95% CI: 0.07 to 0.20, P < 0.00001), with no observed heterogeneity across studies (I^2 = 0%). Conversely, in registry studies, there was no statistically significant difference in dislocation between the two approaches (RR = 0.59, 95% CI: 0.34 to 1.01, P = 0.06),

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although substantial heterogeneity was present within this subgroup (I^2 = 95%).

Study or Subgroup	Experimental Events	Experimental Total	Control Events	Control Total	Weight	Risk Ratio M-H, Random, 95% CI
Barber 1996	0	21	0	28		Not estimable
Castioni 2021	1	72	0	68	2.4%	2.84 [0.12, 68.44]
Fleischman 2019	15	8561	38	2160	13.0%	0.10 [0.05, 0.18]
Gupta 2024	27	2499	29	2255	13.5%	0.84 [0.50, 1.41]
Hermansen 2021	12	1226	1067	29879	13.2%	0.27 [0.16, 0.48]
Hoskins 2020	129	24468	339	65791	15.1%	1.02 [0.84, 1.25]
Jeyaraman 2019	0	47	3	42	2.7%	0.13 [0.01, 2.41]
Liang 2023	0	21	0	20		Not estimable
Ritter 2001	0	130	8	190	2.9%	0.09 [0.00, 1.47]
Rogmark 2022	128	4832	257	3199	15.1%	0.33 [0.27, 0.41]
Skoogh 2019	136	44933	185	51848	15.1%	0.85 [0.68, 1.06]
Tay 2019	1	618	8	870	4.6%	0.18 [0.02, 1.40]
Witzleb 2009	0	30	1	30	2.4%	0.33 [0.01, 7.87]
Xie 2023	0	120	0	201		Not estimable
Total (95% CI)		**87578**		**156581**	**100.0%**	**0.41 [0.24, 0.70]**
Total events	449		1935			

Heterogeneity: Tau² = 0.48; Chi² = 117.65, df = 10 (P < 0.00001); I² = 92%
Test for overall effect: Z = 3.27 (P = 0.001)

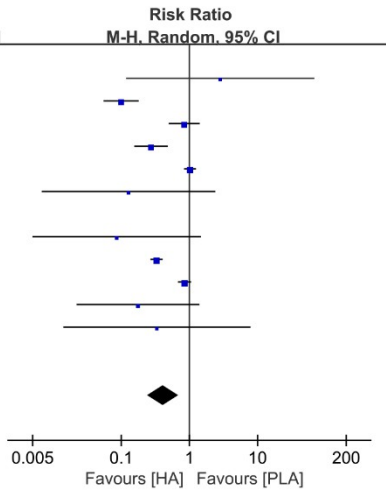

Fig.6 Forest plot for dislocation

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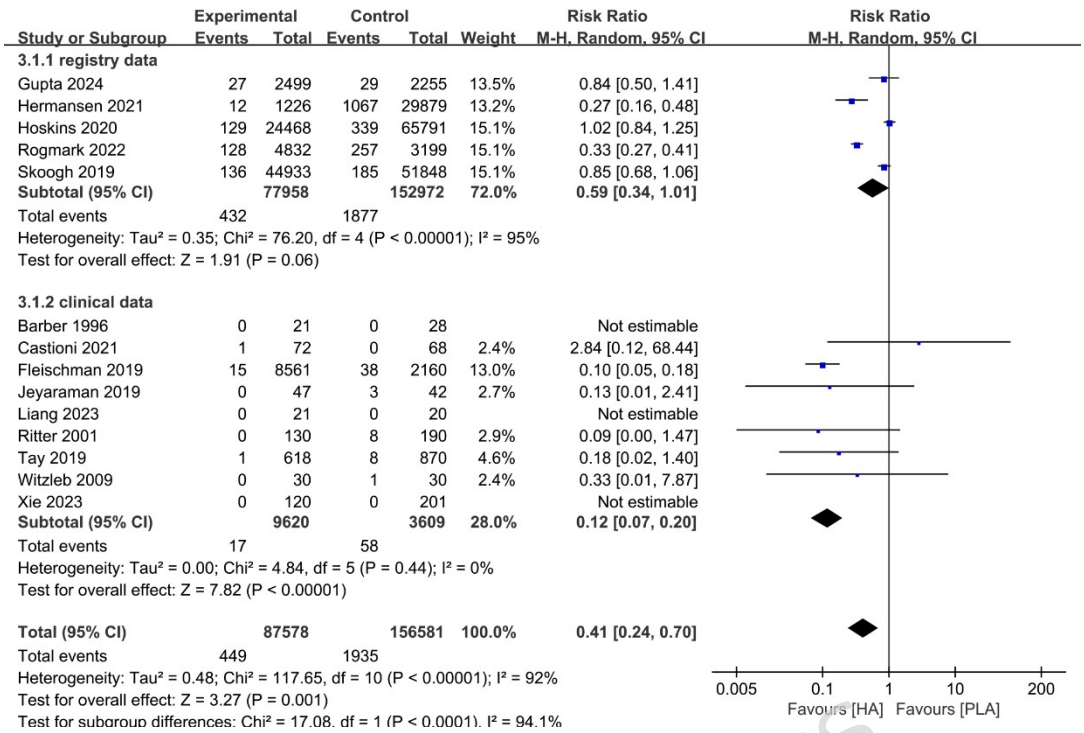

Fig.7 Forest plot for subgroup analysis of dislocation

Infection

Analysis of eight studies [10,13,15,17,22,24,25,28] (n = 193,778) demonstrated that HA was associated with a higher rate of surgical site infection compared to PLA (Fig. 8).

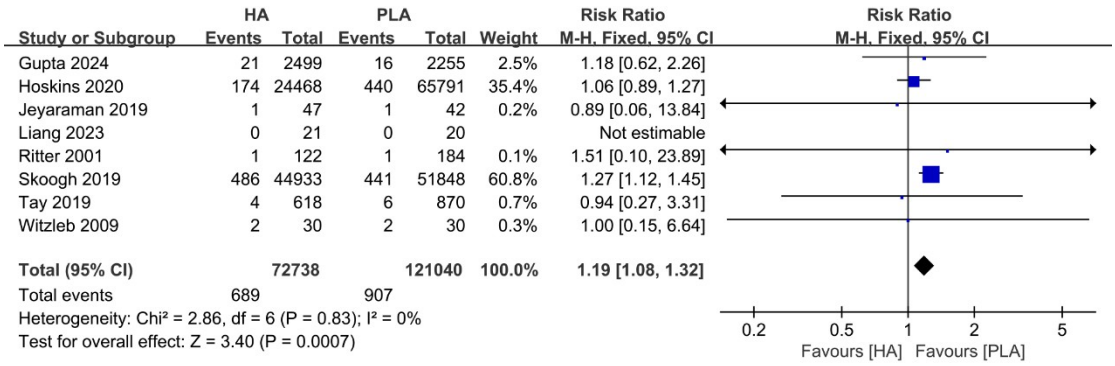

Fig.8 Forest plot for infection

Fracture

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Meta-analysis of six studies [17,22,24,25,28,29] (193,286 participants) found no statistically significant difference in the incidence of fracture between HA and PLA (RR = 0.92, 95% CI 0.78 to 1.08; P = 0.32, Fig.9).

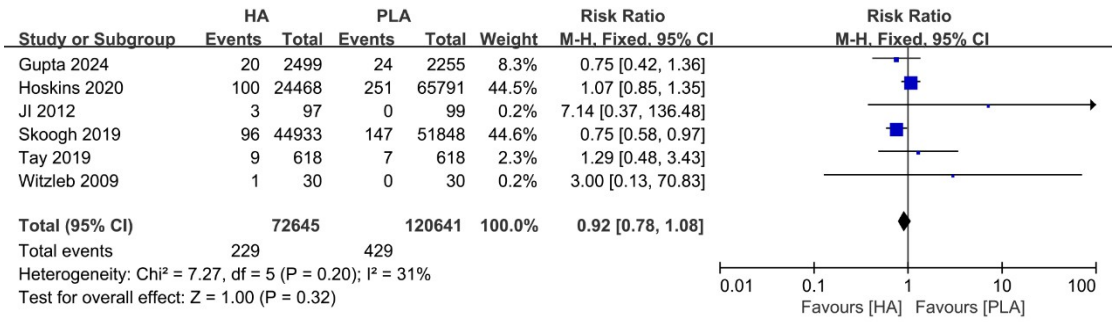

Fig.9 Forest plot for fracture

Loosening

Six studies involving 105,913 participants [10,11,14,15,24,25] demonstrated no clear evidence of a difference in the rate of loosening between the two surgical approaches (RR = 1.51, 95% CI 0.79 to 2.87; P = 0.21; Fig. 10).

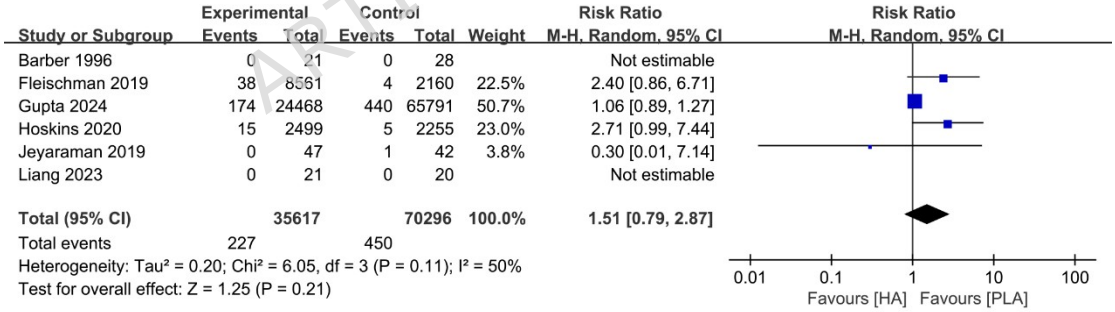

Fig.10 Forest plot for loosening

Heterotopic ossification

HA was associated with a significantly higher incidence of heterotopic ossification compared to PLA, based on four studies involving 457 participants [11,15,16,26] (Fig.11)

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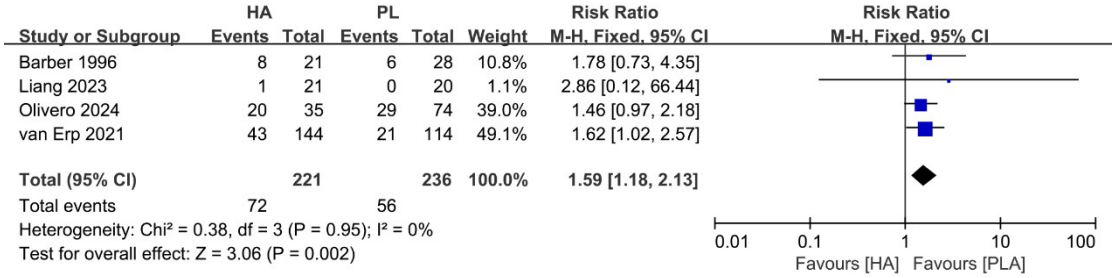

Fig.11 Forest plot for heterotopic ossification

HHS

No statistically significant differences in HHS were found between the surgical groups at any evaluated postoperative time point (>1, >3, >6, and >12 months), based on data from eight studies (Fig. 12)

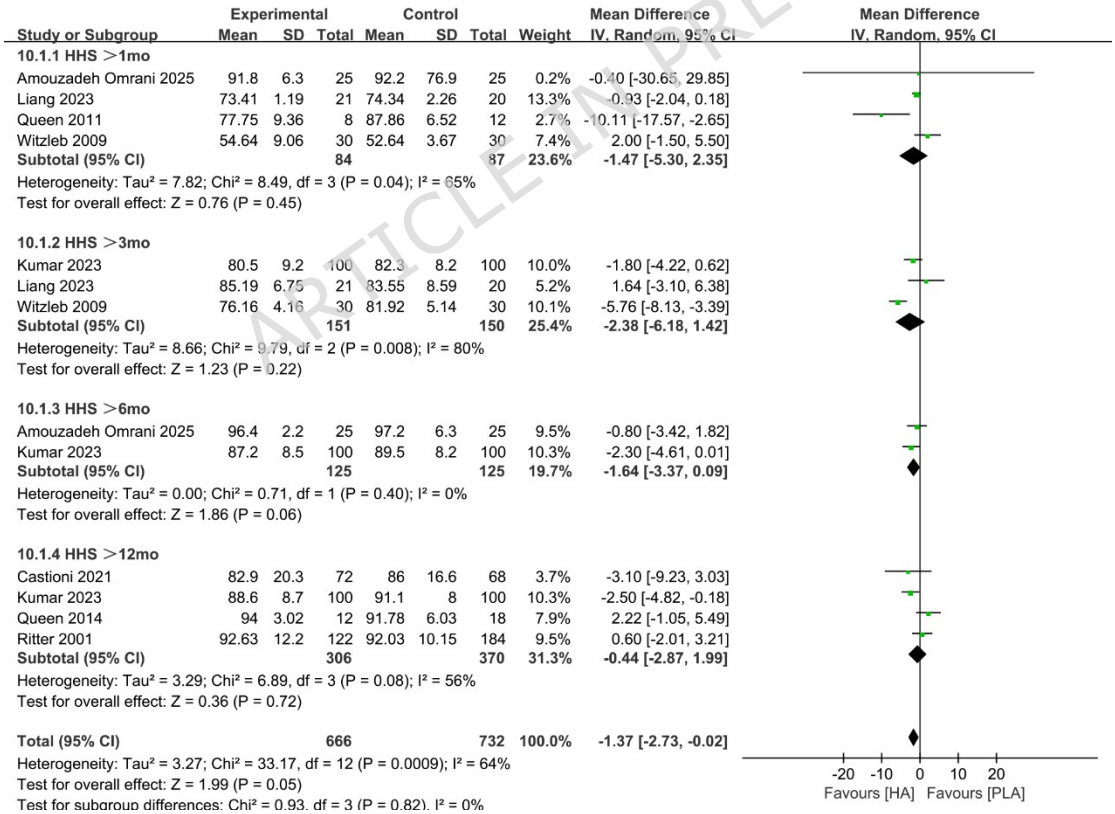

Fig.12 Forest plot for HHS

Abduction angle

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Analysis of abduction angle across four studies [12,15,19,28] (n = 622) found no statistically significant difference between the approaches (MD = -1.52, 95% CI: -4.88 to 1.83; P = 0.37; Fig. 13).

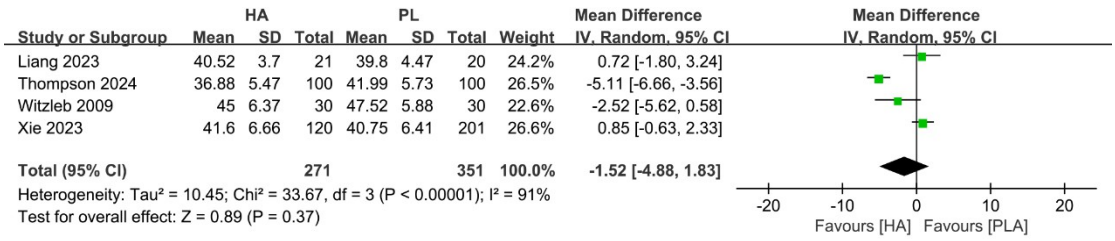

Fig.13 Forest plot for abduction angle

Anteversion angle

Meta-analysis of three studies [12,15,29] (n = 558) revealed no statistically significant difference in the postoperative anteversion angle between HA and PLA (MD = 0.26, 95% CI: -0.38 to 0.90; P = 0.43; Fig. 14).

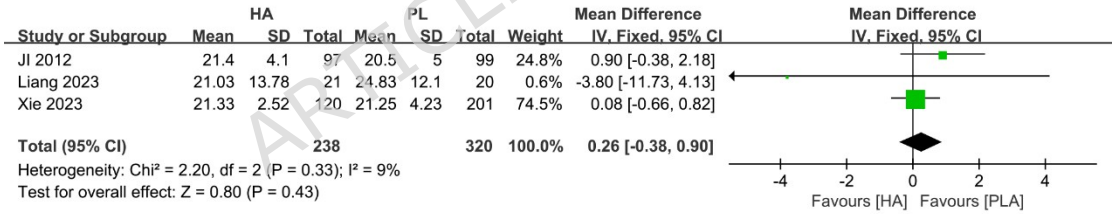

Fig.14 Forest plot for anteversion angle

Discussion

The most frequently criticized aspect of the Hardinge approach is postoperative abductor muscle weakness. Owing to a lack of standard deviation, several retrieved studies on abductor power were excluded from the meta-analysis [36-39]. Manafi Rasi [40] used MRI to compare the abductor muscle diameter before and after employing the Hardinge approach, revealing a reduction from 27.07±7.485 mm to 25.64±7.353 mm. Çakmak [41] employed isokinetic strength testing and gait analysis.

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The results showed that by six months post-surgery, patients had returned to a preoperative gait pattern and had adequate muscle strength. In a rigorously designed RCT, Rosenlund [42] compared gait parameters between the lateral and posterolateral approaches. The results indicated that the posterolateral approach group achieved superior recovery in abductor and flexor power.

Bernasek [43] reported that full weight bearing following surgery significantly reduces the length of hospital stay without increasing the probability of complications or injuries to the abductor muscles. Aavikko [44] used platelet-rich plasma (PRP) to assess its efficacy in promoting the healing of the gluteus medius. The study reported a significantly greater improvement in mean Oxford Hip Score (OHS) and maximal abduction function in the PRP group compared to the placebo group.

While various methods exist to enhance abductor muscle function, surgeons must consider must consider each patient's clinical profile when selecting a surgical approach. Bülbül [45] reported a correlation between morphometric parameters and the presence of Trendelenburg sign, indicating that patients with significantly shorter stature and a higher body mass index exhibited prolonged positivity for Trendelenburg sign. These finding serves as a valuable reference for surgeons when selecting a surgical approach.

HO is a common complication following THA. While surgical techniques have been identified as risk factors, the precise etiology of HO remains unclear. One review highlighted that high-risk factors include hip ankylosis, male sex, and a prior history of HO [46]. Additionally, the use of surgical drainage has been suggested as a potential risk factor. The study conducted by Wei et al. demonstrated that the incidence of HO was markedly higher in the drainage group compared to the non-

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drainage group (32.0% vs. 16.3%) [47]. Soft tissue injury can disrupt the balance of osteoblasts, thereby stimulating the formation of HO [48]. HA causes greater soft tissue damage, which leads to a higher rate of HO [49].

The primary concern associated with the posterolateral approach is the higher dislocation rate, which ranges from 1% to 5% [50]. Repair of the posterior soft tissue may decrease the rate of postoperative dislocation [51]. Kwon [52] performed a meta-analysis that directly compared the posterolateral approach with and without soft tissue repair. The dislocation rates for the repaired and non-repaired groups were 0.49% and 4.46%, respectively. Common methods of repair include transosseous and transmuscular repair [53,54]. Moon [55] performed a prospective randomized controlled trial comparing these two repair methods, reporting that the rate of dislocation was significantly higher in the tendon-to-tendon group. Additionally, complications have been reported after repair, such as avulsion fracture of the greater trochanter [56]. In studies included in our review, Fleischman [14] reported on a total of 2160 patients who underwent THA with soft tissue repair via posterolateral approach. These patients were followed up for two years postoperatively, and the study revealed a dislocation rate of 1.74%.

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In terms of primary outcomes, the PLA group showed better gait velocity and abductor power than the HA group. Although differences in stride length and ground reaction force did not reach statistical significance, trends consistently favored the PLA group. HHS did not differ significantly between groups. Regarding secondary outcomes, the HA group had a lower incidence of dislocation but was associated with higher risks of heterotopic ossification and infection. There were no statistically significant differences between the groups in the rates of fracture or radiographic outcomes.

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Interpretation in Light of Evidence Certainty

The GRADE assessment yielded a 'very low' certainty of evidence for all key outcomes, explicitly assessed due to limitations in study design, substantial heterogeneity, and risk of bias. This does not weaken the findings but crucially improves their interpretability. It provides a transparent framework that prevents overinterpretation of the results, clearly demonstrating that the current evidence is insufficient for definitive conclusions. Thus, the primary value of this review lies in the need for more rigorous, well-designed studies to address these evidence gaps.

Sources of Heterogeneity

Our meta-analysis identified a significant degree of heterogeneity (I^2 = 92%) in the pooled estimate for dislocation risk. As sensitivity analyses did not substantially alter this heterogeneity, we proceeded with pre-specified subgroup analyses, which revealed a critical explanatory factor. Stratification by data source demonstrated a stark and statistically significant difference between subgroups (P for subgroup difference < 0.0001). Among clinical studies, HA was associated with a markedly lower and highly consistent dislocation risk compared to PLA (RR = 0.11, 95% CI: 0.07–0.20; I^2 = 0%). Conversely, in large registry studies representing real-world practice, no statistically significant difference was found between the approaches (RR = 0.59, 95% CI: 0.34–1.01; P = 0.06), although substantial heterogeneity persisted within this subgroup (I^2 = 95%). This divergence suggests that the advantage of HA, evident in clinical studies, is attenuated in routine clinical settings. We postulate this is largely due to the concomitant use of modern stabilizing techniques, such as soft-tissue repair, large femoral heads, and dual-mobility articulations, with PLA in contemporary practice, which effectively modifies its inherent risk profile.

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Limitations of Our Study

Several limitations of this study should be acknowledged when interpreting its results. First, while gait analysis provides a mechanical assessment of postoperative function, the majority of included studies employed a cross-sectional design. This approach inherently limits the ability to control for baseline differences, making pre-operative disparities a potential confounding factor. Second, substantial methodological heterogeneity was observed across studies, including variations in surgical technique, inconsistent follow-up durations, which limits the consistency and interpretability of pooled results. Finally, ensuring complete data collection was challenging, particularly for studies incorporating retrospective data. Although all data were extracted from medical records, some cases lacked adequate follow-up information.

Conclusion

This meta-analysis indicated that PLA had better recovery of gait velocity and abductor power. In terms of dislocation rates, HA presented advantage over PLA. This benefit must be weighed against the higher risks of heterotopic ossification and infection observed with HA. However, these conclusions must be interpreted in the context of important potential confounding factors inherent to the included observational studies. The observed differences in dislocation rates may be influenced by variations in femoral head size, the use of dual-mobility implants, and the specific technique of posterior soft-tissue repair, which was not uniformly reported or adjusted for across studies. Similarly, differences in complication rates such as infection and heterotopic ossification could be affected by indication bias, patient selection, and variations in perioperative protocols. While our analysis synthesizes the best available comparative evidence, residual confounding limits the ability to establish purely

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causal relationships between the surgical approach alone and these outcomes.

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List of abbreviations

THA: Total hip arthroplasty

HA: Hardinge approach

PLA: posterolateral approach

HHS: Harris hip score

BMI: Body Mass Index

RR: risk ratio

CI: confidence interval

MD: mean difference

NOS: Newcastle-Ottawa scale

HO: heterotopic ossification

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GRF: ground reaction force

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Declarations

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and materials

The authors confirm that the data supporting the findings of this study are available within the article [and/or its supplementary materials].

Competing interests

The authors declare that they have no competing interests

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None

Authors' contributions

MXS, and LY collected important background information and data. MXS performed the statistical analyses. MXS and YHW drafted the manuscript. XW, LY, HC, and XFH made substantial revisions to the article. The authors read and approved the final manuscript.

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